REARRANGEMENTS (Rudin Ch. 3)

- Warning: For infinite sums, order matters!

<u>Def</u> Let $\{a_n\}_{n\geq 1}$ be a sequence. We say $\sum_{n=1}^{\infty} \tilde{a}_n$ is a <u>rearrangement</u> of $\sum_{n=1}^{\infty} a_n$ if $\exists\, k: \mathbb{N} \to \mathbb{N}$ bijective s.t. $\tilde{a}_n = a_{k(n)} = a_{k_n}$.

<u>Ex</u> Consider $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$

① Claim: This series converges, and $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} < \frac{5}{6}$.

As $\{\frac{1}{n}\}_{n\geq 1}$ is decreasing and $\lim_{n\to\infty} \frac{1}{n} = 0$, then $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges by the alternating series test, and so $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = -\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ also converges (by HW6#2). This means that the partial sums $s_n = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k}$ converge as $n \to \infty$. When $n = 2m+1$ is odd,

$$s_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots - \frac{1}{2m} + \frac{1}{2m+1}$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \sum_{k=2}^{m} \underbrace{\left(\frac{1}{2k} - \frac{1}{2k+1}\right)}_{0 < \frac{1}{2k(2k+1)} < \frac{1}{4k^2}}$$

As $\sum_{k=2}^{\infty} \frac{1}{4k^2}$ converges (by lecture 16), then $\sum_{k=2}^{\infty} \left(\frac{1}{2k} - \frac{1}{2k+1}\right)$ also converges by the comparison test. So

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \lim_{n\to\infty} s_n = 1 - \frac{1}{2} + \frac{1}{3} - \sum_{k=2}^{\infty} \left(\frac{1}{2k} - \frac{1}{2k+1}\right)$$

$$\underbrace{\qquad\qquad}_{>0}$$

$$< 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

② Claim: The rearrangement $1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \dots$ also converges, and is $> \frac{5}{6}$. This series is given by $\sum_{k=1}^{\infty} \left(\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} \right)$. Note that

$$0 < \frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} = \frac{\cancel{8k^2} - 2k + \cancel{8k^2} - 6k - (\cancel{16k^2} - 16k + 3)}{(4k-3)(4k-1)2k}$$

$$= \frac{8k-3}{(4k-3)(4k-1)2k} < \frac{8k}{k \cdot 3k \cdot 2k} = \frac{4}{3k^2}$$

$$k \geq 1 \Rightarrow \begin{cases} 4k-3 \geq 4k - 3k = k \\ 4k-1 \geq 4k - k = 3k \end{cases}$$

As $\sum_{k=1}^{\infty} \frac{4}{3k^2}$ converges (by Lecture 16), then $\sum_{k=1}^{\infty} \left(\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} \right)$ also converges by the comparison test. Moreover,

$$\sum_{k=1}^{\infty} \left(\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} \right) = 1 + \frac{1}{3} - \frac{1}{2} + \underbrace{\sum_{k=2}^{\infty} \left(\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} \right)}_{>0}$$

$$> 1 + \frac{1}{3} - \frac{1}{2} = \frac{5}{6}.$$

<u>Thm</u> (Riemann's rearrangement result) Let $\sum_{n=1}^{\infty} a_n$ be a series that converges, but does not converge absolutely. For any α, β satisfying $-\infty \leq \alpha \leq \beta \leq \infty$,

there exists a rearrangement $\sum_{n=1}^{\infty} \tilde{a}_n$ so that the partial sums $\tilde{s}_n = \sum_{k=1}^{n} \tilde{a}_k$ obey

$$\liminf_{n\to\infty} \tilde{s}_n = \alpha \quad \text{and} \quad \limsup_{n\to\infty} \tilde{s}_n = \beta.$$

- E.g., this applies to $a_n = \frac{(-1)^{n+1}}{n}$
- Taking $\alpha = \beta$, this says that we can rearrange $\sum a_n$ so that it converges to any real number, or diverges to $+\infty$, or diverges to $-\infty$.

<u>Pf</u>: For $n \geq 1$, set

$$b_n = \begin{cases} a_n & \text{if } a_n \geq 0 \\ 0 & \text{if } a_n < 0 \end{cases} \qquad c_n = \begin{cases} 0 & \text{if } a_n \geq 0 \\ -a_n & \text{if } a_n < 0 \end{cases}$$

Then $b_n, c_n \geq 0 \quad \forall n \geq 1$.

Claim: $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ both diverge. Note

$$\sum_{k=1}^{n} b_k - \sum_{k=1}^{n} c_k = \sum_{k=1}^{n} (b_k - c_k) = \sum_{k=1}^{n} a_k$$

$$\Rightarrow \sum_{k=1}^{n} b_k = \sum_{k=1}^{n} c_k + \underbrace{\sum_{k=1}^{n} a_k}_{\text{converges as } n\to\infty}$$

So: $\sum_{k=1}^{\infty} b_k$ converges $\Leftrightarrow$ $\sum_{k=1}^{\infty} c_k$ converges. If both series converged, then

$$\underbrace{\sum_{k=1}^{n} b_k}_{\text{converge as } n\to\infty} + \underbrace{\sum_{k=1}^{n} c_k}_{\text{converge as } n\to\infty} = \sum_{k=1}^{n} (b_k + c_k) = \underbrace{\sum_{k=1}^{n} |a_k|}_{\Rightarrow \text{ this converges}}$$

which contradicts that $\sum_{n=1}^{\infty} a_n$ does not converge absolutely. So both $\sum_{k=1}^{\infty} b_k$ and $\sum_{k=1}^{\infty} c_k$ must diverge.

As $b_k, c_k \geq 0$ $\forall k \geq 1$, they both diverge to $+\infty$.

Let $B_1, B_2, B_3, \ldots$ denote the nonnegative terms of $\{a_n\}_{n\geq 1}$, in order.

$C_1, C_2, C_3, \ldots$ denote the absolute value of the negative terms, in order.

[Eg.: $\sum a_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots$

b_n: $1 \quad 0 \quad \frac{1}{3} \quad 0 \quad \frac{1}{5} \quad 0 \quad \ldots$

c_n: $0 \quad \frac{1}{2} \quad 0 \quad \frac{1}{4} \quad 0 \quad \frac{1}{6} \quad \ldots$

B_n: $1 \quad \frac{1}{3} \quad \frac{1}{5} \quad \ldots$

C_n: $\frac{1}{2} \quad \frac{1}{4} \quad \frac{1}{6} \quad \ldots$]

Then $\sum_{n=1}^{\infty} B_n$ and $\sum_{n=1}^{\infty} C_n$ differ from $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ only by terms that are zero, so $\sum_{n=1}^{\infty} B_n = +\infty$ and $\sum_{n=1}^{\infty} C_n = +\infty$ as well.

Choose sequences $\{\alpha_n\}_{n\geq 1}$ and $\{\beta_n\}_{n\geq 1}$ s.t.:

- $\alpha_n \to \alpha$ as $n \to \infty$
- $\beta_n \to \beta$ as $n \to \infty$
- $\alpha_n < \beta_n$ $\forall n \geq 1$
- $\beta_1 > 0$

[Eg.: if $\alpha, \beta \in \mathbb{R}$,

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Next, we construct increasing sequences $\{k_n\}_{n\geq 1}$ and $\{j_n\}_{n\geq 1}$ as follows:

① Choose $k_1, j_1 \in \mathbb{N}$ to be the smallest numbers s.t.

$$x_1 = B_1 + B_2 + \ldots + B_{k_1} > \beta_1$$ (this is possible, since $\sum B_n = +\infty$)

$$y_1 = B_1 + B_2 + \ldots + B_{k_1} - C_1 - C_2 - \ldots - C_{j_1} < \alpha_1$$

(this is possible, since $\sum C_n = +\infty$)

② Choose $k_2, j_2 \in \mathbb{N}$ to be the smallest numbers s.t.

$$x_2 = B_1 + \ldots + B_{k_1} - C_1 - \ldots - C_{j_1} + B_{k_1+1} + \ldots + B_{k_2} > \beta_2$$

$$y_2 = B_1 + \ldots + B_{k_1} - C_1 - \ldots - C_{j_1} + B_{k_1+1} + \ldots + B_{k_2} - C_{j_1+1} - \ldots - C_{j_2} < \alpha_2$$

and so on.

By construction we have

$$x_n - B_{k_n} \leq \beta_n \implies \beta_n - B_{k_n} < \beta_n < x_n \leq \beta_n + B_{k_n}$$

$$\implies |x_n - \underbrace{\beta_n}_{\to \beta}| \leq \underbrace{B_{k_n}}_{\text{subsequence of } \{a_n\}_{n\geq 1}} \to 0 \text{ as } n \to \infty$$

$\sum a_n$ conv. $\implies \lim_{n\to\infty} a_n = 0$

$$\implies \lim_{n\to\infty} x_n = \beta$$

(To be continued!)